Rasya Fawwaz

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Technical Skills

Embedded & Firmware: C/C++, FreeRTOS, BLE, SPI, I2C, UART, CAN, STM32, ESP32, nRF52, Raspberry Pi, AWS IoT
Controls & Modeling: MATLAB, Simulink, PID Control, HIL Testing
Hardware & Debug: Oscilloscope, DMM, CANalyzer, KiCad, Altium
Software & Tools: Python, Git, GDB, Embedded Linux, Shell, OpenGCD, CMake SDK Development
HDL: SystemVerilog

Education

University of Washington — Seattle, WA
Bachelors of Science in Electrical and Computer Engineering (GPA: 3.84) — *December 2026*

Relevant Experience

Teaching Assistant — March 2026 – Present
University of Washington, College of Engineering — *Seattle, WA*

- Mentored 40+ students in the design and modeling of complex digital systems using Verilog HDL, resulting in the successful hardware synthesis of intermediate-level projects on Altera DE1-SoC FPGA platforms.
- Diagnosed and resolved real-world timing issues in student-designed sequential and combinational circuits, facilitating a deeper understanding of parasitic elements and signal integrity across multi-clock domains.
- Supervised weekly laboratory sessions, training students to use modern engineering tools including Quartus II, ModelSim, to validate system performance against real-world environmental specifications.

Embedded Engineer — Jan 2026 – Present
PACCAR (Capstone) — *Seattle, WA*

- Developed a motor model in Simulink to perform Hardware-in-the-Loop (HIL) testing prior to physical deployment
- Validated CAN bus signal integrity between the RCM-112 ECU and inverter using CANalyzer and oscilloscopes.
- Designed the system wiring harness, integrating the ECU, inverter, and high-voltage motor connections.
- Architected a PID controller in a team environment to manage a 48V Three-Phase AC Motor with positive and negative torque.

Undergraduate Research Assistant — April 2025 – Sept. 2025
Mobile Intelligence Lab — *Seattle, WA*

- Collaborated within a multi-disciplinary team to develop camera-integrated earbuds over a 5-month lifecycle.
- Engineered a synchronization tool in Python to visualize dual raw camera data streams transmitted via UART.
- Optimized the BLE firmware data pipeline, reducing system latency by 30% to improve wakeword responsiveness.
- Developed Python automation scripts to interface with microprocessors via CLI for streamlined testing and operational validation.

Relevant Projects

Bare-Metal Embedded Suite — Jan 2026 – Present

- Built a from-scratch HAL-free firmware library for the STM32F411, implementing GPIO, I2C, SPI, ADC, and Timer drivers directly from the RM0383 reference manual.
- Designed verification programs per driver to confirm signal integrity and timing correctness at the hardware level.

Self Balancing Robot — Dec 2025 – Feb 2026

- Characterized the dynamics of the robot using Euler Lagrange Equations, simulating the dynamics in Python.
- Applied the PID and State Space controllers in hardware, using IMU values as feedback input to the system
- Tested and created State Space and PID Controllers in simulation, finding kP, kI and kD values to have less than 10% overshoot, a rise time of 0.25s and a steady state error of 0%.

ESP-32 AirTag — June 2025 – Sept. 2025

- Developed a firmware for an ESP32-based tracker, integrating BLE RSSI for distance measurement.
- Authored low-level SPI and I2C drivers from scratch for IMU register configuration and data acquisition.
- Tuned system performance to maintain 70% CPU utilization under peak load, ensuring real-time responsiveness.
- Leveraged FreeRTOS task prioritization to synchronize sensor sampling with wireless communication windows.